

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY CHANGA

M. Tech. (EC) (CSE) Semester – II, Final Examination, 2009-10

EC – 711, RF and Wireless Communication System

Date: 19/06/2010, Saturday Time: 12.30 p.m to 3.30 p.m Maximum Marks:70

Instructions:

1. Attempt all questions
2. Assume suitable and standard data if needed
3. Use of nonprogrammable calculator is allowed

SECTION- I

Q1 (a) A Uniform lossless 100 Ohm line is connected to a load to $50 - j75\Omega$ as shown in figure 1. A Single stub of 100Ω characteristic impedance is connected in parallel at a distance (d_s) from the load. Find the shortest values of d_s and the stub length l_s for a match using Smith Chart. 03

(b) Define, discuss and interrelate the following terminologies

- (i) Group and Phase velocities
- (ii) VSWR and reflection coefficients

04

Q2 (a) Enlist various types of transmission line resonant circuits for $n = 1$ and 2. Explain and compare the short-circuited an open-circuited line resonators for their various parameters. 07

(b) For a lossy line having $V_s = 100$ V, $Z_s = 30 - j40 \Omega$, $Z_o = 50 \Omega$, $l = 10.2\lambda$, $\alpha = 0.01$ Np/ λ , $Z_L = 150 \Omega$; calculate the time-average power dissipated in the line and delivered to the load. 07

OR

Q2 (a) With help of appropriate examples, explain the resistive and reactive L - section matching circuits. 07

(b) A 10-m long, 50Ω lossless transmission line is terminated by a load $Z_L = 100 + j50\Omega$. It is driven by a signal generator that has an open-circuit voltage $V_s = 100$ Volt, $Z_s = 50 \Omega$. The propagating signal has phase velocity of 200 m/microsecond at 26 MHz. Determine the impedance at its input end and the phasor voltages at both ends. 07

Q3 (a) Explain double stub matching network technique. Using graphical approach of Smith chart, find the shortest values of l_1 and l_2 to match the load for the figure 2. 07

(b) A load reflection coefficient is given as $0.4 \angle -30^\circ$. It is desired to get a load with reflection coefficient at $0.2 \angle 45^\circ$. There are two different circuits given in figure 3. However, the information provided for these circuits is incomplete. Complete and verify the designs at 4 GHz. Assume $Z_0 = 50 \Omega$. 07

OR

Q3 (a) Draw and explain the dielectric resonators and design a $TE_{01\delta}$ - mode cylindrical dielectric resonator for use at 35 GHz in the micro strip circuit as shown in figure 4. The substrate height = 0.25 mm and dielectric constant = 9.9. For the material available for DR, the dielectric constant = 36 and the top of DR should have clearance of 1mm from the conducting enclosure. 07

(b) Define, Explain and compare the following terms for series and parallel resonant circuits: (i) Damping Ratio (ii) Input impedance/ admittance (iii) Quality factor in terms of half power frequencies 07

SECTION- II

- Q4 (a) Design a low-pass constant-k T-section that has nominal impedance of 75Ω and a cut-off frequency of 2 MHz. Draw equivalent low-pass constant-k T-section. 03
 (b) Define commensurate lines; with necessary illustration explain 'Richard's Transformation for Filter Synthesis using redundant elements' 04
- Q5 (a) Derive Image impedance using image parameter method. 08
 (b) Explain in detail: Adaptive antenna arrays can benefit the Carrier-to-Interference ratio. 06

OR

- Q5 (a) Compare the protocol model of Signaling System 7 (SS7) with Open System Interconnection protocol model 08
 (b) Discuss microcell application for information superhighway. 06
- Q6 (a) Describe Intelligent microcells with use of necessary sketch and prove that capacity of intelligent microcells is greater than that of AMPS by factor of $7/3$. 08
 (b) Define spectral efficiency. Illustrating necessary examples explain the need of sectorization with necessary sketches. Explain how sectorization can improve spectral efficiency. 06

OR

- Q6 (a) Explain Markov Model for occupied channels in a network with N channels including explanation of effect of call waiting in Markov model. In addition, explain Erlang B formula with mentioning the conditions where Erlang B can be applied. 08
- (b) How many users can be supported for 1% blocking probability if the number of channel is 20? Assume that each user generates 0.1 Erlangs of traffic. Find out peak hour traffic with average duration of call being 2 minutes. The offered traffic intensities for a number of channels are provided in Table 1. 03
- (c) If an SIR of 15 dB is needed for satisfactory forward channel performance in a cellular system, what should be the optimum frequency reuse factor and cluster size if the path loss exponent is $n = 4$. 03

Number of Channels	A
2	0.153
4	0.869
5	1.36
20	12

Table 1: Offered Traffic Intensities

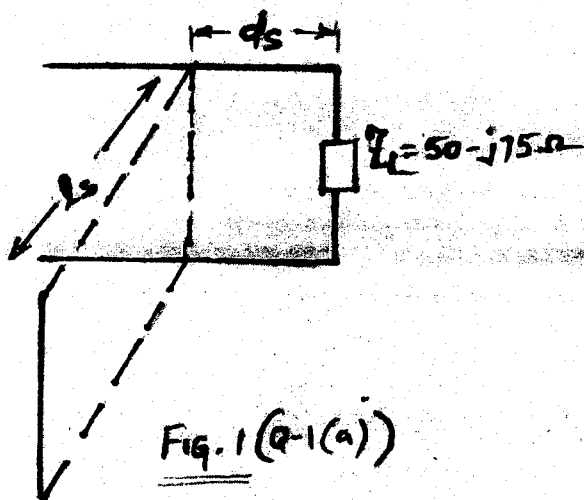


FIG. 1 (Q-1(a))

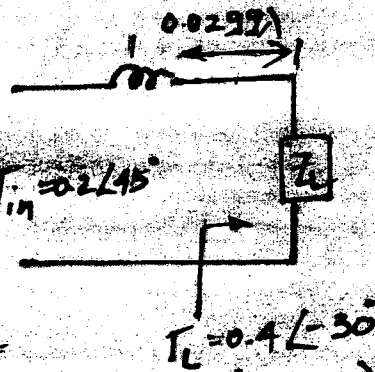
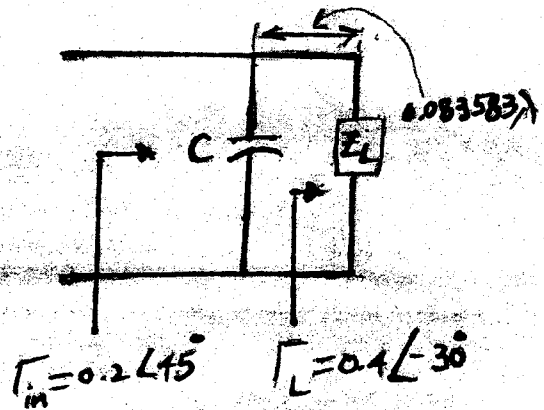


FIG. 3 (Q-3(b))

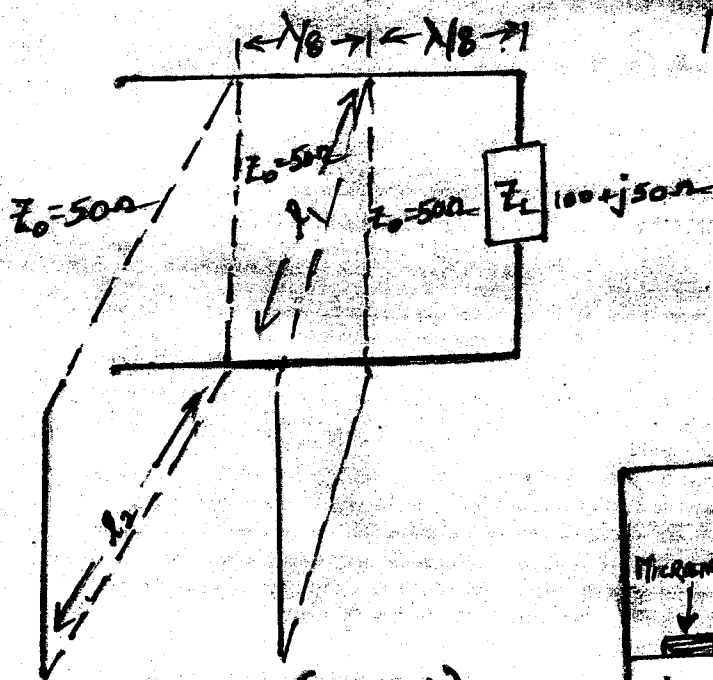


FIG. 2 (Q-3(a))

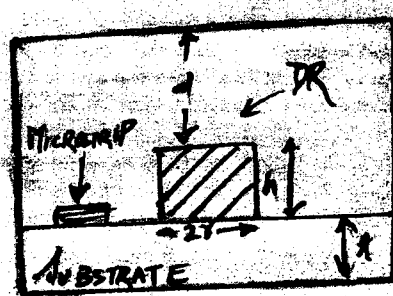


FIG. 4 (Q-3(a))

OPTIONAL